

Designing Interactive Interfaces For E-Learning Apps

‘Study Coven’

Spring 2025

CSBP316 (SEC51)

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April 27, 2025



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Topic Selection and Goal Setting

Objectives

Our goal for our project is to build an interactive website for high school and university students to study together, stay focused, and develop strong academic habits in a safe and supportive online environment.

Project Objectives:

1. Group students into video study calls based on subject, grade, and interests using AI.
2. Monitor group sessions with AI to reduce distractions and encourage focus.
3. Allow students to upload study materials and generate custom study plans with milestones
4. Teach effective study techniques like Pomodoro, mind mapping, and spaced repetition.
5. Encourage students to study in a more healthy and effective way and become more diligent
6. Make the website more engaging with a gamified point system for completing tasks and customizing avatars.
7. Encourage older students to volunteer as peer mentors for younger students.

Website Features

1. AI-Driven Intelligent Grouping

- The system shall automatically assign students to virtual study groups using artificial intelligence.
- Grouping parameters shall include:
 - Academic grade level
 - Subjects of study
 - Responses collected from a brief onboarding survey

2. Focus Monitoring and Distraction Management

- The system shall implement real-time AI-based monitoring during active study sessions.
- The system shall:
 - Detect and identify off-topic discussions outside of designated break periods.
 - Issue subtle focus prompts or reminders for scheduled breaks.
 - Deduct points from participants for repeated or significant distractions.

3. Personalized AI Study Assistant

- The system shall allow users to upload study materials (e.g., PDFs, notes).
- The AI shall:
 - Recommend effective study habits (e.g., hydration reminders, muting notifications).
 - Generate personalized study plans based on user-defined timelines.

4. Study Method Optimization

- The system shall suggest evidence-based study techniques to users.
- Supported techniques shall include, but are not limited to:
 - Pomodoro Technique
 - Access to AI-powered study tools during meetings

5. Integrated Smart Calendar and Goal Management

- The system shall provide a smart calendar tool that allows users to:
 - Track academic deadlines (exams, assignments, projects).
 - Set and customize personal study goals.
 - Receive reminders and task notifications.
 - Track the number of days actively spent studying.

6. Gamification and Leaderboard

- The system shall award points to users based on:
 - Completion of academic and personal objectives.
 - Participation and attendance in study sessions.
 - Providing peer-to-peer assistance or tutoring.
- The accumulated points shall allow users to climb a public leaderboard.

7. Creative Collaboration Spaces

- The system shall provide non-academic collaborative spaces, including:
 - Digital art rooms for drawing and creative expression.
 - Writing rooms for project collaboration or idea sharing.
 - Debate and discussion rooms where users can dynamically group and rate each other's contributions.

8. Shared Music Experience

- The system shall allow study groups to access curated collaborative playlists.
- Users shall have the ability to:
 - Vote to skip or change songs.
 - Suggest and add music tracks in real-time during sessions.

9. Peer Mentorship Program

- The system shall support a mentorship program where older or more experienced students may:
 - Volunteer as study guides for junior students.
 - Participate in group calls in a mentorship/support capacity.

- Earn badges or officially recognized volunteer hours (e.g., Fazaa program certification).

Safety and Moderation Protocols

1. Participant Reporting and Moderation Review

- The system shall allow participants in a study call to initiate a vote to remove disruptive users.
- The system shall automatically log and monitor all removal votes.
- The system shall perform automated reviews to detect patterns of false reporting.
- The system shall support manual moderator review to verify the legitimacy of participant removals if needed.

2. Verified Account Access

- The system shall require all users to sign up and log in using verified institutional credentials
- The system shall prevent access to unverified or anonymous accounts to maintain a secure academic environment.

3. Privacy Assurance

- The system shall have user video cameras disabled by default during all sessions.
- The system shall allow screen sharing functionality during sessions, subject to user permission.
- The system shall protect user data and prevent unauthorized recording of sessions without explicit consent.

User Research

Objective

After extensive requirement gathering, we identified our target users along with their core needs, pain points, and goals.

Target Users

- Students from grade levels 8 to 12, and university.

User Needs

- A safe and secure platform
- Gender-based group separation
- Supportive, judgement-free academic environment
- A distraction-free study experience

- Smart group matching by: Grade level, Subject, Gender

Pain Points

- Struggle to find good study partners
- Difficulty in avoiding distractions while studying
- Hesitation in asking for help due to shame or fear of judgment
- Feeling unsafe or uncomfortable when interacting with strangers online
- Struggle to stay motivated alone

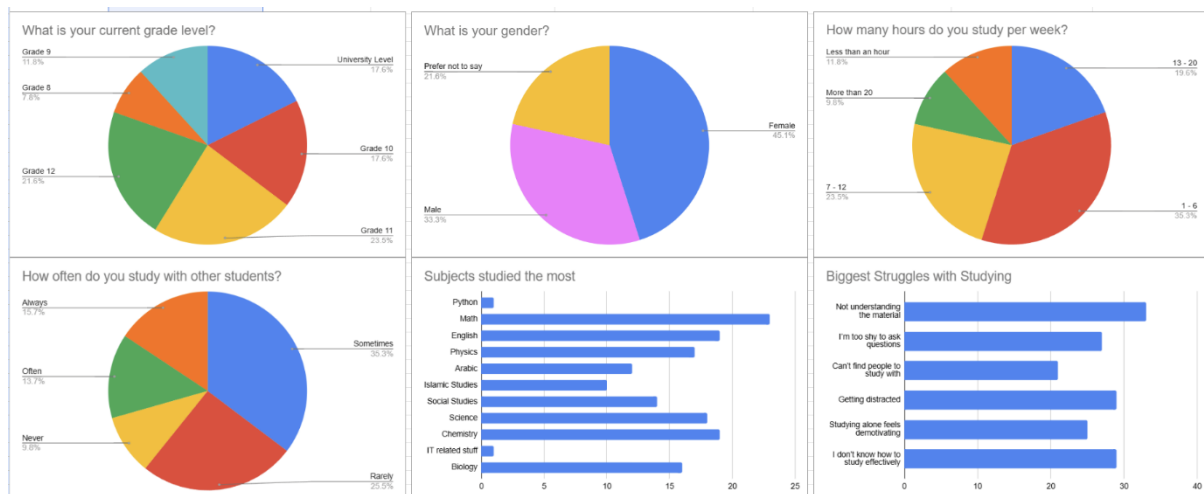
Goals

- Connect with other students
- Study more effectively
- Access tutoring in a supportive environment.
- Study with other students in their own age group and gender group without worrying about dealing with suspicious strangers
- Reduce exam stress

Survey Results

We conducted two surveys; one meant for user requirement gathering and the second created for the purpose of feature validation. The results are as shown below

User Requirement Survey





This survey was created to identify user goals, pain points, and needs.

Grade-level distribution showed that the majority of users are in high school:

- Grade 11 was the most common at 23.5%.
- Grade 12 followed at 21.6%.
- University students and Grade 10 students tied at 17.6% each.
- Grade 9 accounted for 11.8%, while Grade 8 had the smallest group at 7.8%.

Gender distribution leaned heavily female:

- 45.1% identified as female,
- 33.3% identified as male,
- 21.6% preferred not to disclose their gender.

Study hours per week mostly ranged from 1 to 6 hours:

- 35.3% of users studied 1–6 hours weekly.
- The second most common range was 7–12 hours at 23.5%.

Study group frequency indicated that students sometimes study together:

- 35.3% selected "sometimes."
- 25.5% selected "rarely."

Subjects studied were mostly high school core subjects:

- Math was the most common, with approximately 24 responses.
- English and Chemistry tied for second.
- Science, Physics, and Biology followed closely, each receiving over 15 responses.
- Arabic, Social Studies, and Islamic Studies were less common but still notable.

Biggest struggle with studying focused on understanding the material:

- More than 20 responses cited not understanding material as their biggest struggle.
- Getting distracted and not knowing how to study effectively tied for second place.
- 25 users reported that studying alone feels demotivating.

Comfort asking questions in groups showed mixed feelings:

- 27.5% of users reported feeling either neutral or uncomfortable.
- Only 21.6% reported feeling comfortable.

Safety talking to strangers online leaned toward occasional caution:

- 41.2% said they feel safe sometimes.
- 33.3% said they do feel safe.

Use of Emirates ID for secure login was generally well-received:

- 67.9% said they would feel safer using Emirates ID.

Study group gender preference showed a strong same-gender preference:

- 72.2% prefer same-gender study groups.
- 25% reported having no preference.

Motivation to study was mainly driven by personal goals:

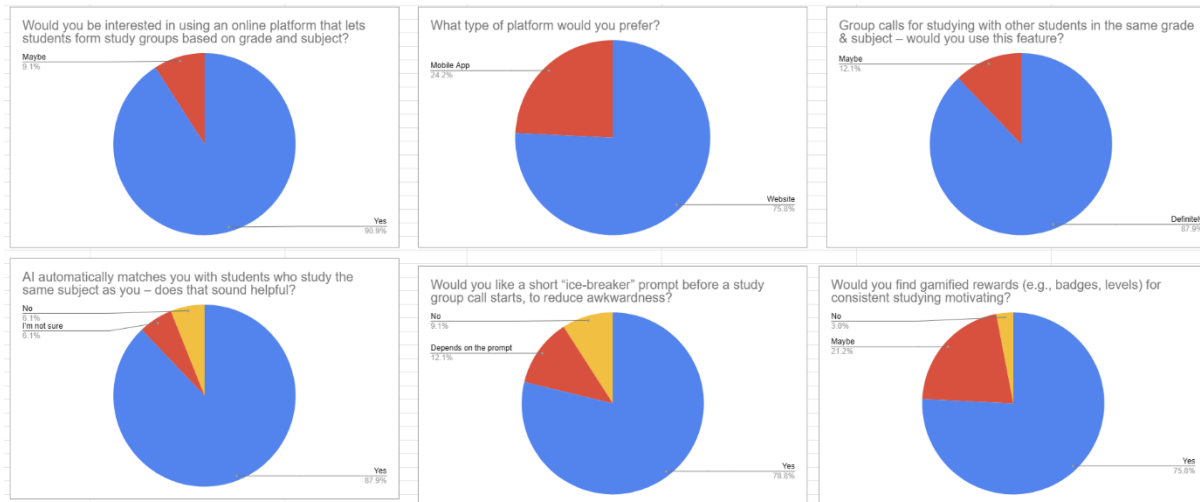
- Over 32 responses cited personal goals as their main motivation.
- Around 30 responses mentioned good grades.
- Group accountability came third.

Interest in tutoring younger students was largely positive:

- 51% would like to volunteer.

- 35.3% would volunteer if it fit their schedule.

Feature Validation Survey



High interest was shown in a platform that groups students based on grade and subject:

- 90.9% of users selected yes.
- 9.1% selected maybe.

Preferred platform for study groups leaned strongly toward a website:

- 75.8% preferred a website.
- 24.2% preferred a mobile app.

Interest in group calls with students of the same grade and subject was high:

- 87.9% selected definitely.
- 12.1% selected maybe.

Use of AI for automatically matching students was viewed positively:

- 87.9% said yes.
- No and I'm not sure both tied at 6.1%.

Ice-breaker prompts before group calls received mostly positive feedback:

- 78.8% selected yes.
- 12.1% said depends on the prompt.

- 9.1% selected no.

Gamified rewards to motivate consistent studying were well-received:

- 75.8% said yes.
- 21.2% said maybe.

Persona

Name: Aysha Al Mehri

Age: 16

Grade Level: Grade 11 (High School)

Gender: Female

Location: UAE

Personality:

Aysha is shy and a little hesitant to speak up in group settings. Asking questions makes her nervous but loves to tutor others.

Study Habits:

- Studies around 7–10 hours a week on her own.
- Prefers small, same-gender study groups where she feels safe and less judged.
- Feels demotivated when studying alone for too long.
- Loves Math, Biology, and Chemistry, and dabbles in Art when she needs a creative break.

Biggest Challenges:

- Struggles with understanding the material.
- Too shy to ask questions during big group discussions.
- Struggles with studying effectively
- Can get easily distracted.

Strengths:

- Exceptional tutor: excels at explaining difficult concepts simply.
- Willing volunteer: happy to mentor younger students, especially in Math and Biology.
- Highly motivated by personal goals, group accountability, and occasional competition.

Feelings about Platform Features:

- Emirates ID login: Makes her feel safer and reassured.
- Same-gender groups: Very important for her comfort.
- **Art as a study subject:** She's curious but would treat it as a relaxing bonus subject.

Ideal Study Group:

A small, focused group of girls who are serious about studying, where asking “dumb” questions

doesn't exist, and support is the unspoken rule. No loud distractions, just mutual respect and encouragement.

Competitive Analysis

While no existing platform offers the full range of features we provide, certain individual features can be found across various other websites.

Competitors

1. Discord + Study Bots

A communication platform with customizable bots and community servers used widely for both casual and study-focused collaboration.

Strengths: There are many different study bots available, and they can be customised to personal preference. The platform also supports text, voice, and screen-sharing capabilities, which enhance collaborative learning. It also does not require any payment to use it.

Weaknesses: The platform is publicly accessible to anyone leading to many hidden dangers for students. In addition to, that the students have to independently search online through different servers to find the one that fits their requirements.

Opportunities: Integrating the personalization and bot functionality of Discord into a closed, education-focused ecosystem would offer students the same flexibility without compromising safety.

2. StudyStream

An online platform offering global study rooms with Pomodoro timers and LoFi music to promote focused, timed study sessions.

Strengths: Users can have meetings with different individuals from around the world while incorporating Pomodoro timers & LoFi music.

Weaknesses: The platform operates on a pay-per-hour model, which may limit accessibility for some users. Additionally, users must either host their own rooms or manually search for and join rooms based on brief descriptions—leading to inconsistent experiences and limited personalization.

Opportunities: In our website we can enhance the user experience by integrating AI-driven room recommendations, automatic group matching, and it's free. In addition to be more safe for the students in the UAE, we require school email account and passwords.

3. StudyTogether

A web-based study community integrated with Discord that provides solo study spaces, timers, goal-setting, and progress tracking.

Strengths: Has a website that is linked to discord where you have solo study rooms. In these rooms you can customize your interface to your personal preference. It also includes a personal timer, YouTube video player, and a place to add your session goals. When signing up they ask what is your education level then assign you to different servers in discord based on that. In addition to having a leaderboard ranking based on your studying hours to motivate people to study. It is also free for anyone who wants to join however some additional features for customization in Discord need to be played for.

Weaknesses: As stated above about Discord, it is a platform that is fully open to the public so it may contain hidden dangers. Users must enter random study rooms or ask around in the public chat for people with their same interest, which is not very efficient.

Opportunities: By integrating these productivity features into a closed, AI-curated environment, the platform could significantly improve both safety and relevance.

4. Focusmate

A virtual coworking platform that pairs users with accountability partners for timed productivity sessions based on selected duration.

Strengths: User can have 25, 50, and 75-minute sessions to finish their daily chores. It comes with different modes such as Desk, Moving, and Quiet Mode. You are put in a room based on which gender you choose to be matched with.

Weaknesses: Does not have an extra feature that helps keep students focused in studying. In addition, room pairing is limited to not only students and is only based on gender.

Opportunities: We can give the students the option to choose the session duration, and they can request to extend it if they want.

5. CaveDay

A structured deep work platform offering facilitated, timed focus sessions led by trained hosts to boost individual productivity.

Strengths: Includes a Planning Workshop feature that allows users to schedule focus sessions in advance, promoting consistency and time management.

Weaknesses: Requires a paid membership, making it less accessible to all users. Additionally, the session schedules follow a fixed structure, offering limited flexibility for those with varying availability.

Opportunities: Introduce student-led weekly workshops where peers can share study strategies, productivity tips, and motivational support. This would foster community engagement and provide practical, relatable advice from fellow learners.

Design Ideation

Sketch:

Dashboard

StudyCoven.com

Welcome back, Sara! (23) Ready to lock in?

Studied hours this week: Current streak: '3'

Summary of what you have achieved

Daily Quote: ~~~~~

My account
!! Report Problems
Settings

Study Goals

StudyCoven.com

April

Sun	Mon	Tue	Wed	Thu	Fri	Sat

Not Yet started | in progress | Done | On Hold

My account
!! Report Problems
Settings

Study Session

StudyCoven.com

What Are You Studying Today?

Pick

☒ Science ☒ Calc ☒ Python → Picks

Time Spending

☐ Not sure ☐ 60 minutes
☐ 1 hour ☐ 3 hour
☐ 2 hour ☐ 4 hour

Tutor wanted: yes ☐ No ☐

Feeling chatty?
☐ Yes ☐ No

Studying Technique for today session

☐ Pomodoro
☐ ...
☐ ...

How Many People?

☐ 2
☐ 3
☐ 4
☐ Other

Creating Study Session

My account
!! Report Problems
Settings

Searching

StudyCoven.com

2/4

Searching for students.....

Tutor

StudyCoven.com

Subject to tutor

choose

☒ Python ☒ Java

Available Time

☐ 1 hour ☐ 3 hour
☐ 2 hour ☐ 4 hour

More Questions

Searching for students

My account
!! Report Problems
Settings

Meeting

StudyCoven.com

30:27
 ⏱️ time left to call

📄 upload study documents

🔊 mic

🖥️ share screen

💬 message

🔍 has more function

📞 close call

🧠 AI feature

📌 See Breaker Pull Out 2. Divide study materials Tutor

Leader Board

← → ↻ WWW.StudyCoven.com ★ :

StudyCoven

Dashboard Study Goals Study Session Tutor AI Leader Board My account Report Problems Settings

Overall Today Week Month Friends

1 2 3 4 5 6 7 8 9 10 11 12 13

level

123 120 115 110 105 100 95 90 85 80 75 70 65 60 55 50 45 40 35 30 25 20 15 10 5 0

AI

AI

← → ↻ WWW.StudyCoven.com ★ :

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Dashboard Study Goals Study Session Tutor AI Leader Board My account Report Problems Settings

AI bot

user

AI bot

user

Enter Prompt

My account

← → ↻ WWW.StudyCoven.com ★ :

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My Account

First Name Last Name

Grade level School/University

School email Phone number

Personal Questions

Which one are you?

☐ Introvert ☐ Extrovert ☐ Omnivert

Are you easily distracted?

☐ Yes ☐ No

Are you interested in these activities?

☐ Art ☐ Debate ☐ Writing

See more Questions ↓

edit Info

Report Problems

← → ↻ WWW.StudyCoven.com ★ :

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Dashboard Study Goals Study Session Tutor AI Leader Board My account Report Problems Settings

Report Problem

Choose Problem

Unit description

max 500 word

Submit

Setting

← → ↻ WWW.StudyCoven.com ★ :

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Dashboard Study Goals Study Session Tutor AI Leader Board My account Report Problems Settings

Setting

change Password

old Password New Password

Submit

Language setting

English

Dark theme

Text size

Deactivate account

Prototyping Design

Link for Version 1:

<https://www.figma.com/proto/1dTkdSDc0jkuN0KWY4uFO/Give-up--me-?node-id=653-17646&t=sNeOPSI49WfnQkoT-1&scaling=min-zoom&content-scaling=fixed&page-id=645%3A7592&starting-point-node-id=653%3A19022>

Link of Project file:

<https://www.figma.com/design/1dTkdSDc0jkuN0KWY4uFO/Study-Coven?node-id=653-17646&t=OqIH8NWzTojiFX4X-1>

Usability Testing

Round 1 of testing:

During the first round of user testing, several minor bugs were identified. All reported issues were patched before moving on to the second round of testing, which focused on the design and ease of use of the prototype.

Round 2 of testing face to face with university and high school students:

In the second round, users found the app's design easy to learn and navigate. Approximately 90% were able to use the app independently without any assistance or guidance, highlighting its user-friendly design. Users also commented that the interactivity of the features was enjoyable and expressed strong interest in testing a fully developed version in the future.

Design Complaints

- Users disliked the plain white background on the sign-up and sign-in pages.
- Some felt that the homepage appeared too empty and suggested adding more descriptions about the app's features.
- A few users mentioned that certain colour choices made some text difficult to read.
- The calendar page size was suggested to be fixed.
- In the meeting finding the AI App was difficulty since it was white
- The messages button was difficult to use and find

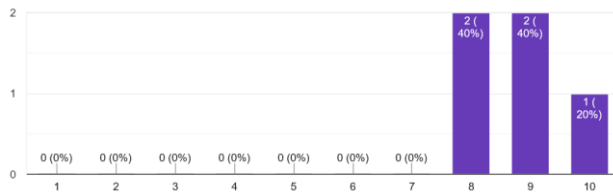
Area of Improvement

1. Will adjust the colours to be more engaging and readable.
2. The home page will have an extra description added to it
3. Calendar page sizes will be adjusted
4. AI App colour in the meeting will be changed to stand out more
5. Messages button was added to the meeting down bar to make it easier to find

Round 2 of testing, Link sent with a google form to fill to different WhatsApp groups:

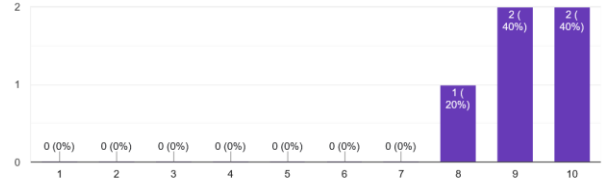
Rate how easy to navigate the website from 1-10.
5 responses

Copy chart



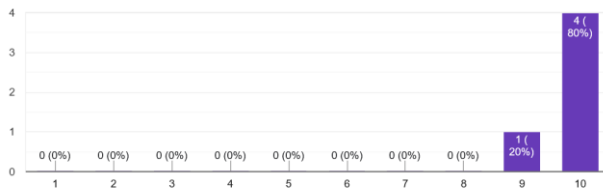
Rate the ease of learning of how to learn to use this website from 1-10.
5 responses

Copy chart



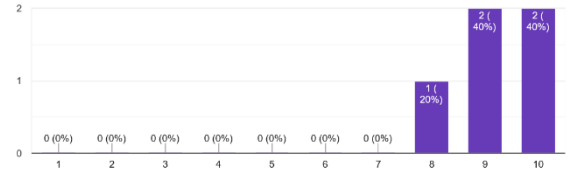
Rate the services provided in the website from 1-10.
5 responses

Copy chart



Rate the language used in the website from 1-10.
5 responses

Copy chart



Iteration and Re-design

Version 2: <https://www.figma.com/proto/1dTkdSDc0jkuN0KWY4uFO/Give-up--me-?node-id=653-24878&t=H5Dydy7jMPA8LUP1-1&scaling=min-zoom&content-scaling=fixed&page-id=653%3A24597&starting-point-node-id=653%3A26254>

Testing was done again to check that the problems user commented on was fixed.

Conclusion

No project is ever perfect but given the time frame we had and our lack of prior experience with Figma, this project successfully achieved almost all our goals. We applied what we learned in HCI across all phases of the project from gathering requirements to sketching prototypes, and finally, creating a design that is as accessible and user-friendly as possible.

We incorporated symbols throughout the app to aid navigation by relying more on recognition rather than recall. The settings app shows that there will be a feature to adjust text sizes for those who need it, and we offer voice functionality in the AI assistant to accommodate a wider range of users.

The most important thing we focused on was following the Usability Design Principles, which we succeeded in, as shown by how users were able to navigate the website without any problems. In addition to functionality, we focused on making the website visually appealing to enhance the overall user experience.

Regarding Figma: we encountered persistent, unpredictable bugs where elements would randomly appear, disappear, or even change size without reason. If any issues arise during use, refreshing the Figma link usually resolves them—though, frankly, sometimes it felt like Figma had a mind of its own.